Chemistry - Graduate Level Assessment 12 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. A student made three successiveweighingsof an object as follows: 9.17 g, 9.15 g, and 9.20 g. What is the average weight of the object in milligrams?

   (a) 9180 mg
   (b) 9150 mg
   (c) 9200 mg
   (d) 9135 mg
   (e) 9155 mg

2. Calculate the ratio of energy loss when a 4 Mevalpha particle travelling through carbon collides with nuclei (scattering) to the energy loss when it collides with bound electrons.

   (a) $2.5 \times 10^{-}3$

   (a) $5.5 \times 10^{-}3$

   (a) $1.5 \times 10^{-}3$

   (a) $1.0 \times 10^{-}3$

   (a) $1.2 \times 10^{-}3$

3. The dissociation sequence of thepolyproticacid H_3 PO_4 showsthreeBronsted-Lowry acids. Rank them in order of decreasingstrengths.

   (a) H_2 PO_4$^-$, HPO_4^2-, H_3 PO_4
   (b) HPO_4$^-$2, H_3PO_4, H_2PO_4$^-$
   (b) HPO_4^2-, H_2 PO_4$^-$, H_3PO_4
   (b) H_2 PO_4$^-$, H_3PO_4, PO_4^3-
   (c) PO_4^3-, HPO_4^2-, H_2 PO_4$^-$

4. What is the molarity of ethanol in 90-proof whiskey? Density of alcohol = 0.8 g/ml.

   (a) 9.20 M
   (b) 12.10 M
   (c) 5.60 M
   (d) 7.83 M
   (e) 2.95 M

5. When air is let out of an automobile tire, the air feels cold to the touch. Explain why you feel this cooling sensation.

   (a) The air feels cold because it mixes with colder atmospheric air
   (b) The air feels cold due to a chemical reaction within the tire
   (c) The air temperature falls because it is doing work and expending energy
   (d) The cooling sensation is a result of the air's rapid expansion
   (e) The cooling sensation is due to the air's humidity

True / False (10 marks)

Answer True or False

6. True or False: The magnitude of the spin angular momentum of a proton is $9.13 \times 10^{\{} - 35\} J \mathring{u} s$.

6. True or False: The pH of a 0.25 M solution of Na_2CO_3 is neutral and measures around 7.00.

7. True or False: After 58 seconds, 0.0125 g of the starting material will remain from a 0.0500 g sample in a first-order reaction with k = 86 h^-1.

7. True or False: The decomposition of 4.90 g of KClO 3 resulted in a weight loss of 0.384 g, indicating that 20% of the original KClO 3 decomposed.

8. True or False: To prepare a 0.25 molal solution with 293 g of NaCl, 20 kg of water is required.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the mass of ice that can be melted at $0^u singtheheatreleasedfromcondensing100gofsteama$

11. Discuss the mass of 1 liter of carbon monoxide (CO) at standard temperature and pressure (STP), including the underlying principles of gas behavior and calculations involved.

End of Paper